ALGEBRA 1 · FOUNDATIONS OF ALGEBRA

Algebra Tiles Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Algebra Tiles Lab** on Mr. K's Classroom: Algebra 1 → Foundations of Algebra → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Can 3x + 2 be simplified to 5x? Explain your thinking.

2. What do you think happens when you put a +x tile and a –x tile together?

2 Explore use the tool and record what you see

Combine like terms

Build each expression on the mat, remove the zero pairs, then record what is left.

Expression you build	Zero pairs removed	Simplified
2x + 3x		
4x + 2 – x		
3 – 5		
x^2 + 2x – 3x + 4		
–2x + 5 + 2x – 1		

Distributive property

Use the Distributive property tab. Count every tile in the rectangle.

Expression	x-tiles	1-tiles (sign)	Simplified
2(x + 3)			
3(x – 2)			
4(x + 1)			
5(x – 3)			

3 Write the rule fill in each blank from what you observed

1. Like terms have the same _____ raised to the same _____.

2. A positive tile and its matching negative tile make a _____, which is worth _____.

3. To combine like terms, add or subtract their _____ and keep the variable part the same.

4. $a(x + c) =$ _____ + _____, because there are a rows and each row holds one x and c ones.

4 Check no tool: show your work

1. Simplify $7x - 2 + 3x + 9$.

2. Simplify $-4y + 6 - y - 6$.

3. Simplify $3(x - 4)$.

4. Simplify $2x^2 + 5x - x^2 - 8x$.

5 Reflect

A classmate says $2x + 3x^2 = 5x^3$. Use tiles to explain the mistake.